MARINDUQUE STATE UNIVERSITY

College of Information and Computing Sciences

Department of Information Technology

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DELIVERABLE #7

G-TRAMS

GASAN TRICYCLE REGULATORY AND AUTOMATED MANAGEMENT SYSTEM

USER'S MANUAL OF THE SYSTEM

Municipal Client:	Municipality of Gasan, Marinduque
Municipal Mayor:	HON. LIDANY A. LAO-BALDO
Authorizing Bodies:	Sangguniang Bayan • MTFRB • BPLO
System Version:	v2.4.0 (Production Release)
Deployment URL:	https://g-trams-official.vercel.app
Capstone Group:	Group 5 (CICS - MarSU)
Lead Backend Architect:	Jhon Kyn Axix H. Cabrigas
Frontend Lead & UI/UX:	Justine S. Lachica
Quality Assurance Lead:	Jay Vincent G. Motol
DevOps & Cloud Engineer:	Ian Simon Francis M. Rey

Pamahalaang Bayan ng Gasan • Sangguniang Bayan • MTFRB

Published for Academic Year 2026–2027 • Approved for Municipal Implementation

DOCUMENT METADATA & REVISION HISTORY

VERSION	RELEASE DATE	AUTHORS / CONTRIBUTORS	SUMMARY OF OPERATIONAL MILESTONES
v1.0.0	September 14, 2026	Group 5 (CICS)	Initial baseline for Proposal Defense; preliminary system architecture and wireframe workflows.
v1.8.0	October 18, 2026	Justine S. Lachica, Jay Vincent G. Motol	Midterm release incorporating multi-role authentication, 5-step MTOP wizard, and BPLO queue.
v2.2.0	November 2, 2026	Jhon Kyn Axix H. Cabrigas, Ian Simon Francis M. Rey	Gemini Vision AI OCR parsing, COA physical OR booklet serialization, and Sunday 12MN backups.
v2.4.0	November 7, 2026	Group 5 (CICS)	Final production release for Final Defense and LGU Turn-over; comprehensive 6-role manual and 17 figure catalog.

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1. SYSTEM OVERVIEW & CITIZEN'S CHARTER

The **Gasan Tricycle Regulatory and Automated Management System (G-TRAMS)** is an enterprise-grade municipal e-governance platform commissioned by the Local Government Unit (LGU) of Gasan, Province of Marinduque. Developed by Marinduque State University - College of Information and Computing Sciences (MarSU CICS) BSIT Capstone Group 5, G-TRAMS automates the regulatory lifecycle of motorized tricycles-for-hire across the municipality.

1.1 Municipal Mandate & Legal Framework

- **Republic Act No. 7160 (Local Government Code of 1991):** Section 447 (a)(3)(vi) delegates the franchising and regulation of tricycles-for-hire to the Sangguniang Bayan and Municipal Mayor via the Municipal Tricycle Franchising and Regulatory Board (MTFRB).
- **Municipal Franchising Ordinance Section 4 (Fleet Quota Limit):** Enforces a statutory individual operator ceiling of **maximum two (2) motorized tricycle units** to eliminate commercial monopoly, prevent unauthorized subleasing, and ensure equitable livelihood distribution among Gasan transport operators.
- **Republic Act No. 11032 (Ease of Doing Business & ARTA):** Mandates digital citizen service delivery with strict Service Level Agreements (SLAs), completing simple evaluations within 3 working days and complex regulatory audits within 7 working days.

- **Republic Act No. 10173 (Data Privacy Act of 2012):** Strictly enforces **Data Minimization**; sensitive biometric markers (such as blood type) are deliberately excluded from AI OCR extraction routines, recording only legally mandated licensing attributes.
- **Commission on Audit (COA) Guidelines:** Precludes unverified online payments; fees are collected strictly via Over-the-Counter Cashier operations and bound to physical serialized Official Receipt booklets (`OR-GASAN-2026-XXXX`).

1.2 The 5-Step Application Lifecycle

G-TRAMS establishes an immutable, five-stage Citizen's Charter workflow:
Step 1: Online Filing (Operator files application, vehicle specs, and documents via 5-Step Wizard; AI OCR pre-check) $ ightarrow$ **Step 2: BPLO Scrutiny** (Dual-Pane inspection, OCR discrepancy check, and physical inspection scheduling) $ ightarrow$ **Step 3: Treasury Settlement** (Cashier collects ₱500 fee, logs serialized receipt `OR-GASAN-2026-XXXX`) $ ightarrow$ **Step 4: Mayoral Sign-Off** (Hon. Lidany A. Lao-Baldo & MTFRB authorize permit) $ ightarrow$ **Step 5: Physical Release** (Operator claims sticker; instant watermarked e-Copy with dynamic QR activates on mobile device).

1.3 Geographic Scope & Route Zones

The territorial jurisdiction of G-TRAMS covers all **25 official Barangays** of the Municipality of Gasan, categorized into three (3) distinct operational route zones:

- **Central Zone (Poblacion Town Proper Loop):** Barangay I (Poblacion), Barangay II (Poblacion), Barangay III (Poblacion). Commercial center, municipal hall, public market, and port.
- **North Zone (Northern Coastal & Upland Route):** Bahi, Bangbang, Cabugao, Mangiliol, Masiga, Tapuyan, Tiguion. Northern agriculture, fishing barangays, and circum-island road link.
- **South Zone (Southern Coastal & Interior Valley Route):** Antipolo, Bachao Ibaba, Bachao Ilaya, Bacong-Bacong, Banot, Banuyo, Bognuyan, Dawis, Dili, Libtangin, Mahunig, Matandang Gasan, Pangi, Pinggan, Tabionan.

1.4 Transport Associations Directory (17 Accredited TODAs)

Every commercial tricycle unit in Gasan must affiliate with an accredited transport association: BATODA, POB TODA, GT TODA, LUG TODA, NBI TODA, BANGBANG IPIL TODA, TAB TODA, TIGUION TODA, MASIGA TODA, MAT TODA, PG TODA, 4B TODA, DPAB TODA, ILAYA TODA, GASAN CENTRAL TODA, BAHI TODA, and NON-TODA (monitored municipal independent commercial units).



Figure 1: G-TRAMS Public Citizen Landing Page & Municipal Portal — Features the top navigation bar with Gasan municipal seal, live statistics counter (Registered Operators, Accredited TODAs, Active Franchises), dual Call-to-Action buttons ("LOG IN" and "REGISTER"), municipal mandate announcement, and 5-step Citizen's Charter interactive service cards.

2. SHARED AUTHENTICATION & SECURITY INFRASTRUCTURE

G-TRAMS implements a centralized, role-based authentication gateway that guarantees high security while ensuring accessibility for transport workers and administrative personnel alike.

2.1 Multi-Identifier Login Architecture

To accommodate operators without active email addresses, the login portal (`/login`) accepts either an **11-digit Philippine mobile phone number** (e.g. `09123456789`) or a **verified email address**. Credentials are cryptographically salt-hashed using bcrypt, and authenticated sessions are governed by JSON Web Tokens (JWT).

Procedure 2.1: Portal Sign-In

Purpose: Authenticate authorized users and route them to their assigned role dashboard.

1. Navigate to `https://g-trams-official.vercel.app/login`.
2. Enter your registered 11-digit mobile number or email address.
3. Type your account password. Click the eye toggle glyph to check visibility if needed.
4. Click the maroon **"Log In"** button.

✓ Expected Result: Session token is issued and user is immediately directed to their role console.

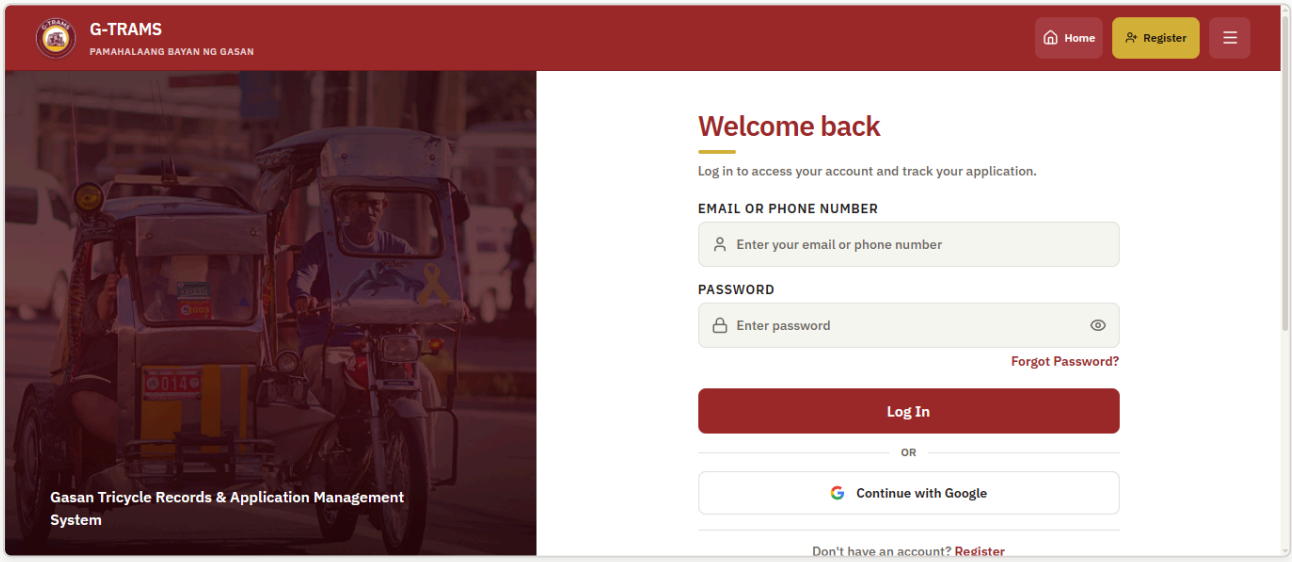


Figure 2: Shared Multi-Identifier Authentication Login Terminal — Secure login portal accessible via /login supporting both 11-digit Philippine mobile numbers (09XXXXXXXXX) and verified email addresses, password field with visibility toggle, Google OAuth single-sign-on integration, and direct link to password recovery.

2.2 Account Registration with Secondary Recovery Contact

To prevent permanent account lockout when mobile SIM cards are damaged or lost, the registration workflow enforces the collection of a mandatory secondary recovery contact (backup mobile phone or email).

Procedure 2.2: New Account Registration

Purpose: Onboard new tricycle operators and transport leaders with account recovery safeguards.

1. Click **"Register"** on the portal header.
2. Input your Full Legal Name matching official government credentials.
3. Enter your Primary Mobile Contact and a mandatory Secondary Recovery Contact.
4. Select your official Barangay Residency from Gasan's 25 official barangays.
5. Select your affiliated TODA Association and input your unit body number.
6. Configure a password (minimum 8 characters with uppercase and numeric digit).
7. Click **"Create Account"** and enter the 6-digit OTP code dispatched to your contact.

✓ Expected Result: Account is created in MongoDB Atlas and user is transitioned to their active garage.

Figure 3: Citizen & Operator Account Registration Interface — Multi-field onboarding console collecting legal applicant name, primary mobile contact, mandatory secondary recovery contact (backup mobile/email) to prevent lockouts, residency selection from Gasan's 25 official barangays, accredited TODA affiliation, and initial body number specification.

2.3 Multi-Channel Password Recovery via 6-Digit OTP

If credentials are lost, operators can request a 6-digit One-Time PIN (OTP) valid for five (5) minutes dispatched via SMS or email to either their primary or secondary recovery contact.

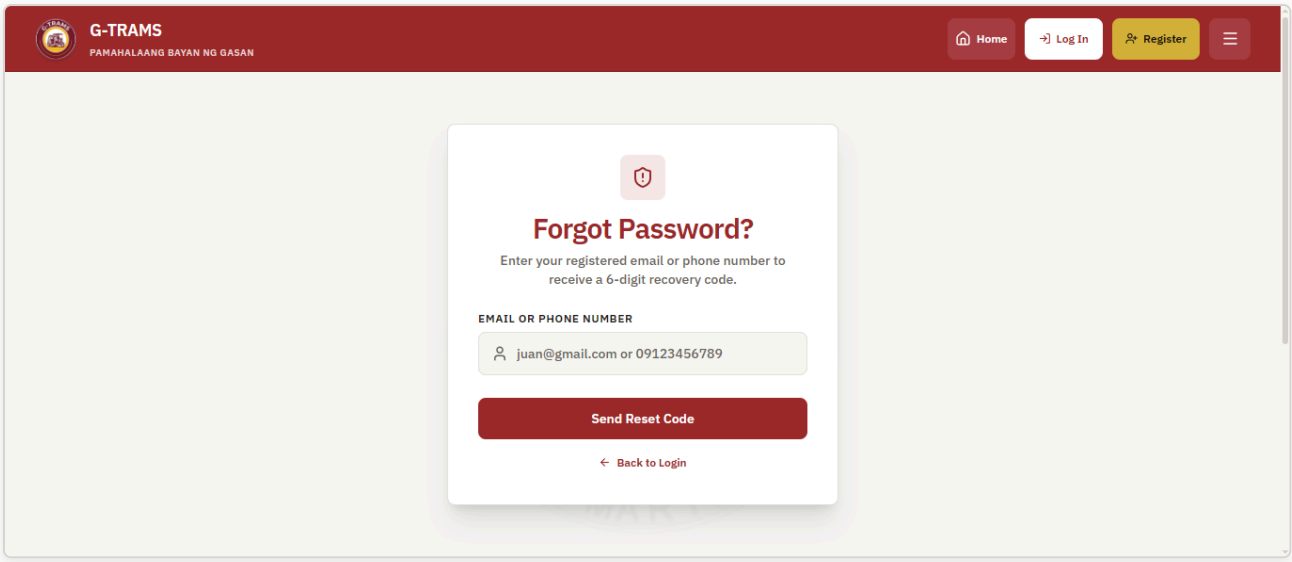


Figure 4: Multi-Channel Password Recovery & OTP Verification Modal — Security modal displaying the red alert shield, contact input field (juan@gmail.com or 09123456789), 6-digit one-time PIN (OTP) verification mechanism with 5-minute timeout window, and password reset form.

2.4 Role-Based Access Control (RBAC) Governance Matrix

CAPABILITY / ROUTE	PUBLIC	OPERATOR	TODA PRES	BPLO OFFICER	CASHIER	SYS ADMIN
Citizen's Charter & Announcements	Read	Read	Read	Read	Read	Read
Roadside QR Verification (`/verify/:id`)	Read	Read	Read	Read	Read	Read
5-Step Online MTOP Wizard	❌	Full	Full	❌	❌	❌
TODA Roster Digitization & OCR	❌	❌	Full	Review	❌	❌
Dual-Pane Document Review & Approvals	❌	❌	❌	Full	❌	❌
Treasury Payment Terminal & OR Logging	❌	❌	❌	❌	Full	❌
COA Daily Remittance CSV Export	❌	❌	❌	❌	Full	❌
IT Telemetry, DRP Restore & Policy	❌	❌	❌	❌	❌	Full

Strict Separation of Duties (SoD) Guardrail

System Administrators are restricted to IT infrastructure and cannot review or approve franchises. Cashiers are restricted to fee receipting and cannot alter permit details or reject applications.

3. PUBLIC VISITOR & PASSENGER GUIDE

G-TRAMS provides transparent public services accessible to all citizens, tourists, and law enforcement officers without requiring account registration or login.

3.1 Public Portal Navigation & Announcements

Citizens navigating to `https://g-trams-official.vercel.app/` can review real-time transport telemetry (Registered Operators, Accredited TODAs, Active Franchises), official municipal bulletins, and traffic advisories published by the Sangguniang Bayan.

3.2 Roadside Mobile Camera QR Verification (`/verify/:id`)

Every franchised tricycle in Gasan features an official QR code sticker affixed to the sidecar windshield and driver digital ID. Any smartphone camera can scan this code to independently verify vehicle legality in real time.

Procedure 3.1: Roadside QR Inspection

Purpose: Verify legitimate commercial tricycle franchises and detect colorum units roadside in under 3 seconds.

1. Open your smartphone camera or any QR scanner application.
2. Aim the camera at the physical QR decal on the tricycle windshield or driver digital ID.
3. Tap the recognized URL link: `https://g-trams-official.vercel.app/verify/:id`.
4. Inspect the official LGU verification card:
   - **Validity Badge:** Green (VALID LGU GASAN FRANCHISE) or Red (EXPIRED / REVOKED).
   - **Operator Profile:** Full Legal Name, photo, and registered contact.
   - **Vehicle Specifications:** Plate Number, Motor Number, Chassis Number, Make, and Body Number.
   - **Association & Route:** Accredited TODA, assigned Route Zone, and permit expiration date.

✓ Expected Result: Official municipal verification renders immediately, confirming genuine registration records.

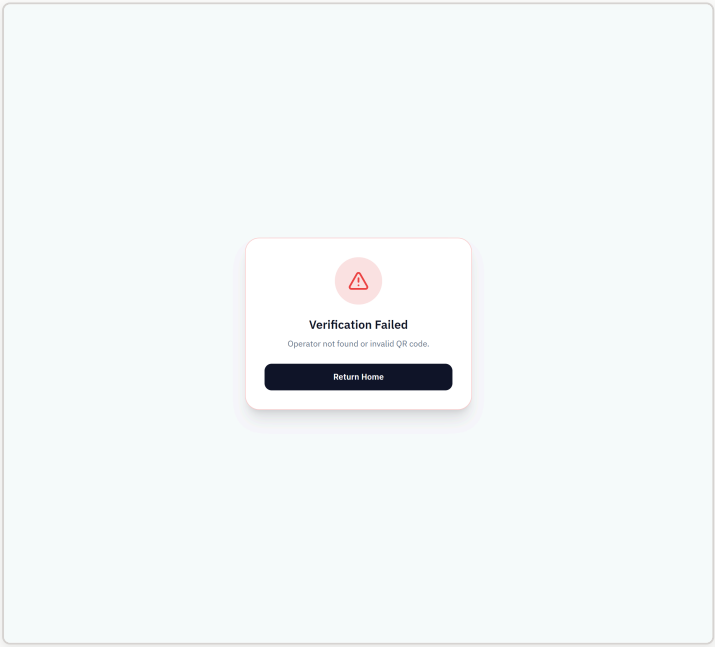


Figure 5: Public Roadside Mobile QR Verification Terminal (/verify/:id) — Passenger and traffic enforcer mobile verification screen displaying real-time franchise validity badge (Active / Under Warning / Expired / Suspended), operator legal name, verified tricycle photograph, LTO plate number, engine/motor number, chassis number, assigned body number, and authorized route zone.

3.3 Official Municipal Fare & Tariff Matrix Calculation

Pursuant to Sangguniang Bayan Municipal Fare Ordinances, tricycle fares in Gasan are standardized:

- **Base Regular Fare: ₱15.00** for the first kilometer within municipal routes.
- **Succeeding Kilometer Rate: ₱2.50** for each succeeding kilometer traveled.
- **Mandatory 20% Discount (RA 11364):** Students, Senior Citizens, and Persons with Disability (PWD) are entitled to a 20% statutory discount: $$ ext{Discounted Fare} = ext{Regular Fare} imes 0.80$$

> **Passenger Protection & Fare Complaint Hotline**
>
> In case of fare overcharging or refusal to convey passengers, report the body number and plate number to the BPLO Helpdesk at (042) 342-1234 or email support@gasan.ph.

4. TRICYCLE OPERATOR GUIDE

The Operator Portal empowers commercial tricycle operators to manage their fleet, file online franchise applications, and access tamper-evident digital credentials.

4.1 Operator Garage Dashboard & Fleet Quota Limit (Max 2 Units)

Upon signing in, operators access their central garage dashboard:

- **Fleet Quota Progress Bar (Section 4 Enforcement):** Enforces the statutory limit of **maximum two (2) units** per individual: `Unit Capacity 1 / 2 — 1 Slot(s) Available` (or `Unit Capacity 2 / 2 — Quota Limit Reached`). Attempting to file a 3rd unit hard-locks the application form with an ordinance alert.
- **Lifecycle Status Tracker:** Tracks milestones: `Applied` $ ightarrow$ `Pending Review` $ ightarrow$ `For Payment` $ ightarrow$ `For Signing` $ ightarrow$ `Ready for Pickup` $ ightarrow$ `Active`.
- **"Fix Issues" Direct Resolution:** Red discrepancy alerts allow operators to click "Fix Issues", jumping straight to the erroneous step with misread data highlighted in yellow.

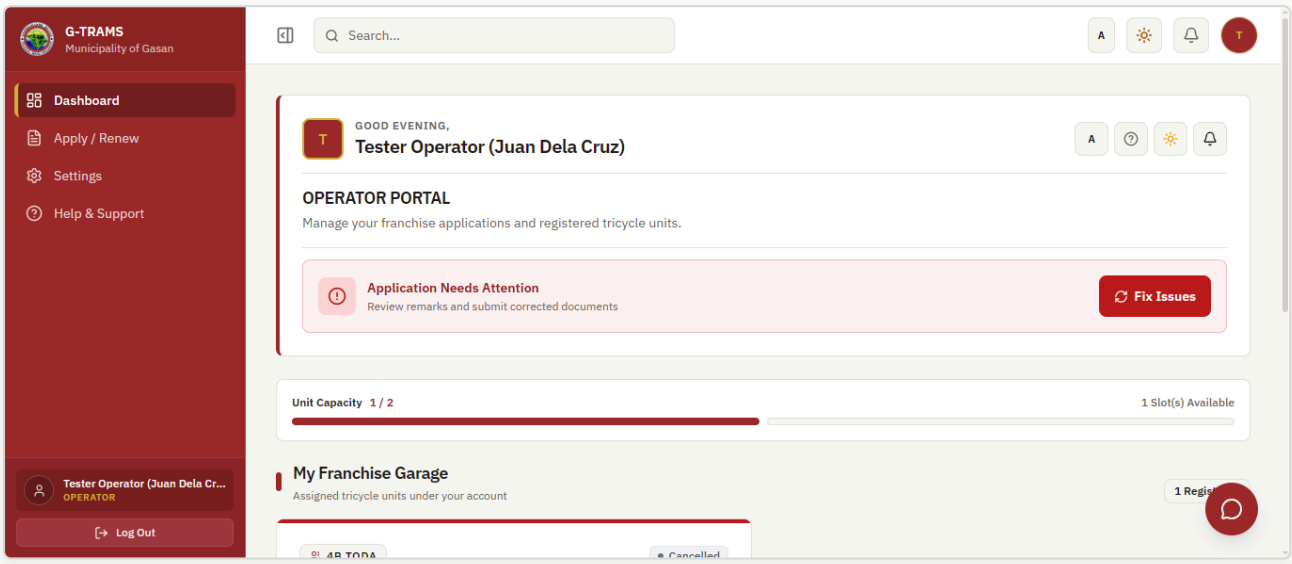


Figure 6: Tricycle Operator Main Garage Dashboard — Comprehensive operator fleet management console displaying the Section 4 statutory quota progress bar ("Unit Capacity 1 / 2 - 1 Slot(s) Available"), active unit status badges, application lifecycle tracker (Needs Attention alert with direct Fix Issues button), digital ID shortcut, and floating municipal help widget.

4.2 5-Step Online MTOP Application Wizard

Procedure 4.1: Filing an Online MTOP Application

Purpose: Apply for New Franchise, Annual Renewal, or Transfer/Dropping without municipal queueing.

1. Click **"Apply for Franchise"** from the Operator Dashboard.
2. **Step 1 (Operator & Driver):** Confirm legal name and barangay address. Toggle Operator-is-Driver or input hired driver credentials with Driver's License photo and expiry date.
3. **Step 2 (Vehicle Specs):** Enter LTO Plate, Engine, Chassis numbers, Make/Model, and Year. The system executes duplicate prevention queries.
4. **Step 3 (TODA & Barangay Clearances):** Select affiliated TODA (green verified badge renders if matched in roster) and upload Barangay Clearance.
5. **Step 4 (Cedula / CTC):** Enter current fiscal year Cedula serial number, issue date, and upload document photo.
6. **Step 5 (Review & Affidavit):** Review summary, confirm sworn affidavit under Article 172 of the Revised Penal Code, and submit. Interrupted sessions are saved via `localforage` draft autosave.

✓ Expected Result: Application transitions to "Pending Review" with a reference ID.

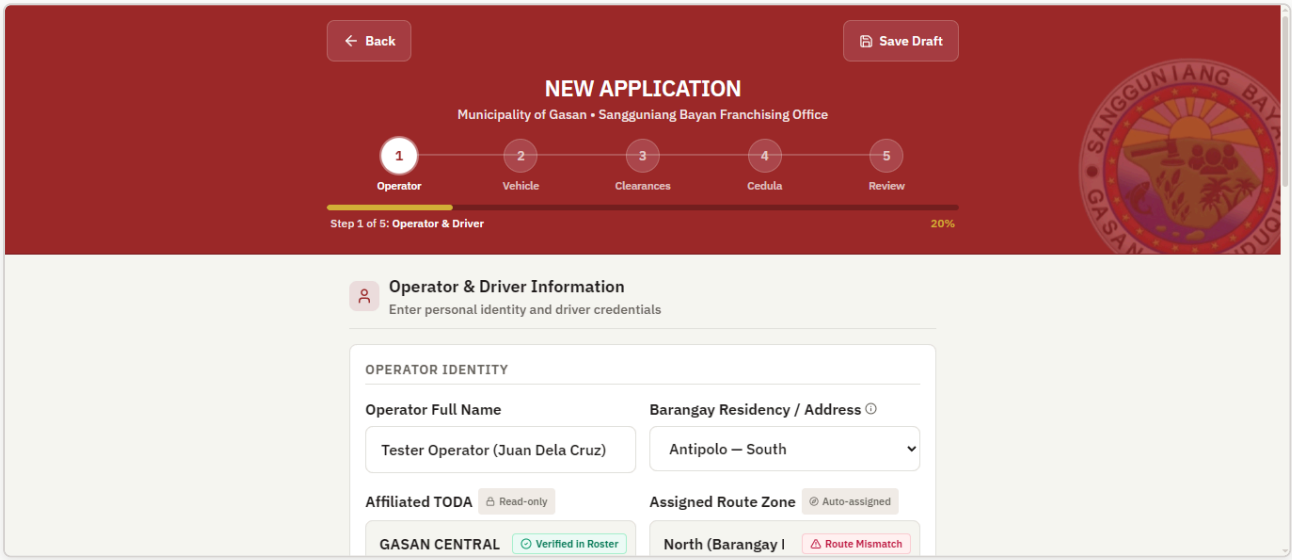


Figure 7: 5-Step Online MTOP Application Wizard (Operator & Route Validation) — Multi-step online franchise filing interface showing 20% progress tracker (Step 1 of 5: Operator & Driver), applicant profile fields, barangay residency dropdown, TODA verification badge (✓ Verified in Roster), and route conflict detector (⚠ Route Mismatch warning).

4.3 Multimodal AI OCR Document Intake & Dual LTO Segregation

G-TRAMS features multimodal Google Gemini Vision OCR (`gemini-3.8-flash`) to parse uploaded credentials and automatically populate technical fields:

- **Dual LTO Segregation:** Dedicated separate upload slots for the **Certificate of Registration (CR)** (ownership and technical specifications) and the latest **Official Receipt (OR)** (annual LTO roadworthiness payment).
- **Pre-Check Validation:** Blurry or cut-off documents trigger instant confidence warnings prompting re-capture before final submission.

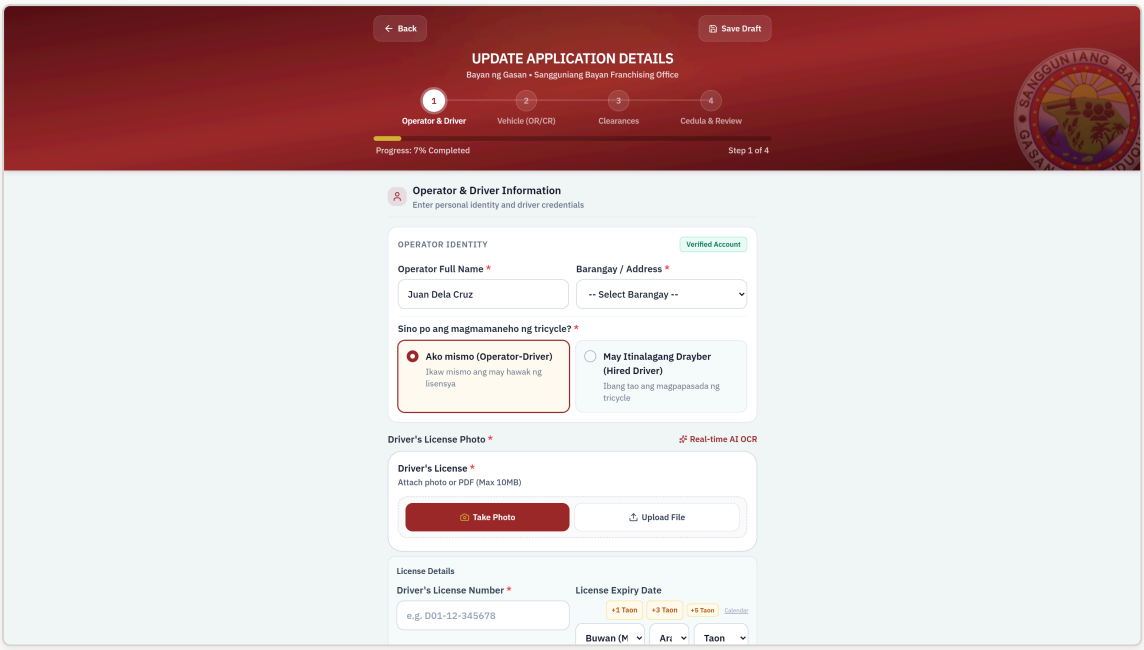


Figure 8: Multimodal AI OCR Document Intake & LTO Credential Segregation — Dedicated upload slots for LTO Certificate of Registration (CR), LTO Official Receipt (OR), Driver's License (with RA 10173 data minimization), Barangay Clearance, and Community Tax Certificate (Cedula), highlighting real-time Gemini Vision OCR text parsing and automated field pre-population.

4.4 Data Minimization Compliance (RA 10173)

Under the Data Privacy Act of 2012 (RA 10173), non-essential biometric markers such as **blood type** and medical indicators are strictly omitted from OCR extraction schemas. The system captures only Driver's License Number, Expiry, DL restriction codes (Code A/A1), and birthdate.

4.5 Digital MTOP Certificate & Watermarked e-Copy Retrieval

When an application reaches `Ready for Pickup` or `Active`, clicking **"View E-Copy"** renders the high-resolution digital permit featuring the diagonal security watermark:
`"APPROVED - ELECTRONIC COPY • VALID ISSUANCE"`
and dynamic QR verification code along with Mayor Hon. Lidany A. Lao-Baldo's official signature.

4.6 Voluntary Franchise Cancellation & Support Ticketing

Operators relinquishing commercial operations can click **"Request Voluntary Cancellation"**, specify grounds (Vehicle Sold, Out of Service, Transfer of Residence), and route their request for dropping certification. The integrated Help & Support module enables submitting inquiries directly to municipal IT staff.

5. TODA PRESIDENT GUIDE

TODA Presidents maintain accredited transport association rosters, issue digital clearances, and enforce collision-free body number allocations.

5.1 Association Member Directory & Status Verification

Navigating to `/submit-members` renders the association management console. The header displays association metrics (e.g. *GASAN CENTRAL TODA*, 89 Members, 2 Active Units). The directory provides real-time search by driver name, body number, and plate number.

5.2 Gemini Vision AI OCR Digitization of Paper Rosters

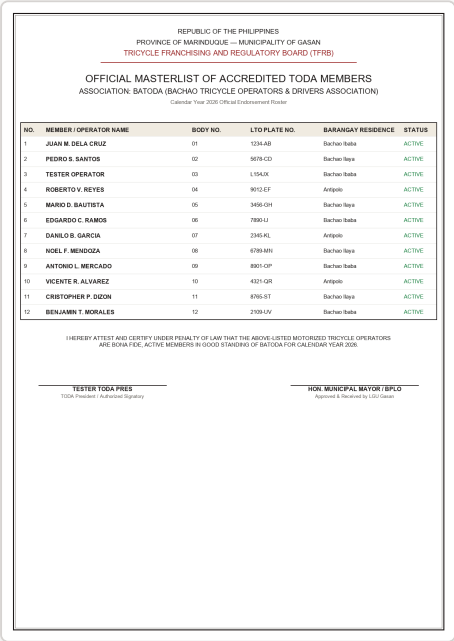
To eliminate manual typing of historical paper member logbooks, TODA Presidents can upload physical document scans for automated AI transcription.

Procedure 5.1: Digitizing Paper Rosters

Purpose: Convert physical association paper masterlists into certified database records via AI OCR.

1. Select **Tab 2: "Upload Roster Document"** in the TODA portal.
2. Upload a clear photo or scan of your physical paper membership sheet.
3. Click **"Process Roster with AI OCR"**. Gemini Vision parses tabular rows into Operator Name, Body Number, and Plate Number.
4. Review extracted rows in the editable staging table; correct typos or click "+ Add Member" to append rows.
5. Click **"Submit Certified Roster to BPLO"** to commit the roster to the municipal masterlist.

✓ Expected Result: Member roster is accredited and indexed for instant validation during operator application filing.



REPUBLIC OF THE PHILIPPINES
PROVINCE OF MARINDUQUE — MUNICIPALITY OF GASAN
TRICYCLE FRANCHISING AND REGULATORY BOARD (TFRB)

OFFICIAL MASTERLIST OF ACCREDITED TODA MEMBERS
ASSOCIATION: BATODA (BACHAO TRICYCLE OPERATORS & DRIVERS ASSOCIATION)
Calendar Year 2026 Official Endorsement Roster

NO.	MEMBER / OPERATOR NAME	BODY NO.	LTO PLATE NO.	BARANGAY RESIDENCE	STATUS
1	JUAN M. DELA CRUZ	01	1234-AB	Bachao Ibaba	ACTIVE
2	PEDRO S. SANTOS	02	5678-CD	Bachao Ilaya	ACTIVE
3	TESTER OPERATOR	03	L154JX	Bachao Ibaba	ACTIVE
4	ROBERTO V. REYES	04	9012-EF	Antipolo	ACTIVE
5	MARIO D. BAUTISTA	05	3456-GH	Bachao Ilaya	ACTIVE
6	EDGARDO C. RAMOS	06	7890-IJ	Bachao Ibaba	ACTIVE
7	DANILO B. GARCIA	07	2345-KL	Antipolo	ACTIVE
8	NOEL F. MENDOZA	08	6789-MN	Bachao Ilaya	ACTIVE
9	ANTONIO L. MERCADO	09	8901-OP	Bachao Ibaba	ACTIVE
10	VICENTE R. ALVAREZ	10	4321-QR	Antipolo	ACTIVE
11	CRISTOPHER P. DIZON	11	8765-ST	Bachao Ilaya	ACTIVE
12	BENJAMIN T. MORALES	12	2109-UV	Bachao Ibaba	ACTIVE

I HEREBY ATTEST AND CERTIFY UNDER PENALTY OF LAW THAT THE ABOVE-LISTED MOTORIZED TRICYCLE OPERATORS ARE BONA FIDE, ACTIVE MEMBERS IN GOOD STANDING OF BATODA FOR CALENDAR YEAR 2026.

TESTER TODA PRES
TODA President / Authorized Signatory

HON. MUNICIPAL MAYOR / BPLO
Approved & Received by LGU Gasan

Figure 9: Physical TODA Association Member Masterlist Scanned Document — Authenticated paper roster from Gasan Central TODA utilized as input for Gemini Vision AI OCR digitization, detailing operator names, unit body numbers, and plate registrations.

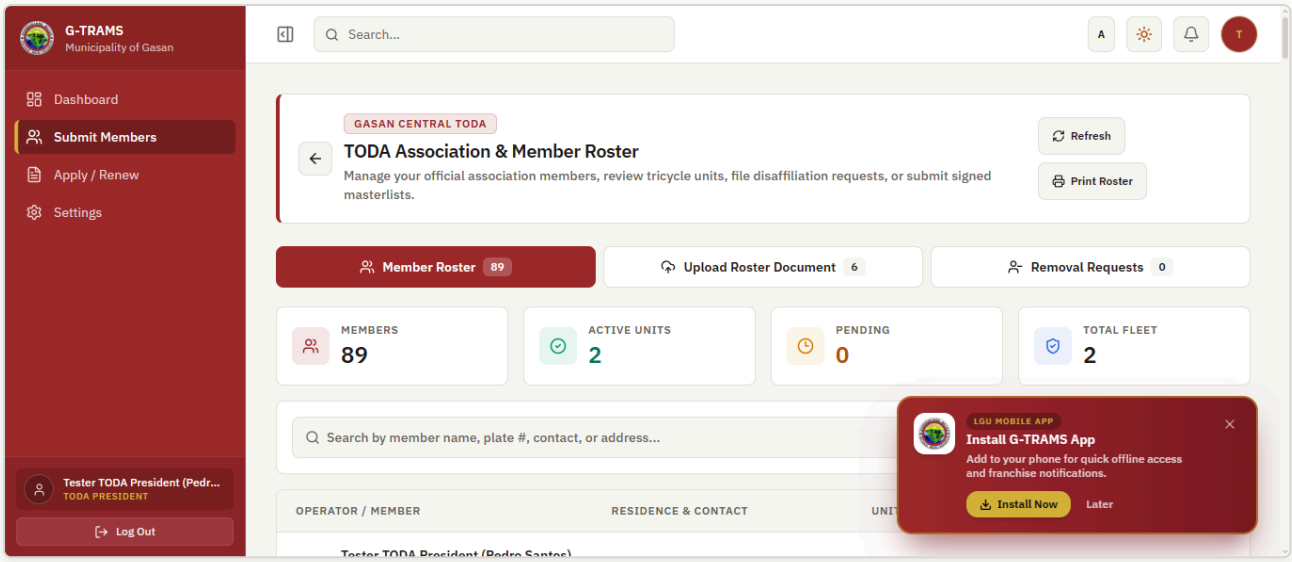


Figure 10: TODA President Association Member Management Terminal — Association administration console displaying active member count (89), active commercial units (2), multi-tab layout ("Member Roster", "Upload Roster Document", "Removal Requests"), member search filter, and PWA installation prompt.

5.3 Association Body Number Collision Prevention

G-TRAMS enforces unique compound database indexing (`todaAssociation + bodyNumber`). Attempting to assign an existing body number triggers an immediate collision block:
`"Duplicate Body Number Detected: Unit #X is already registered to [Driver Name]. Collision prevented."`

5.4 Member Delisting & Removal Requests

Under Tab 3 ("Removal Requests"), presidents can petition BPLO to delist inactive or non-compliant drivers by selecting standard grounds (Relocated, Bylaw Violation, Non-payment of dues, Resigned).

6. BPLO LICENSING OFFICER GUIDE

The Business Permits and Licensing Office (BPLO) serves as the primary municipal regulatory authority, evaluating applications, inspecting vehicles, and enforcing traffic sanctions.

6.1 Operations Dashboard & Pipeline Telemetry

The BPLO Dashboard (`/bplo-dashboard`) displays live executive KPIs: Active Franchises, Pending Approvals, In Pipeline, Expiring Units, and the TODA fleet distribution chart.

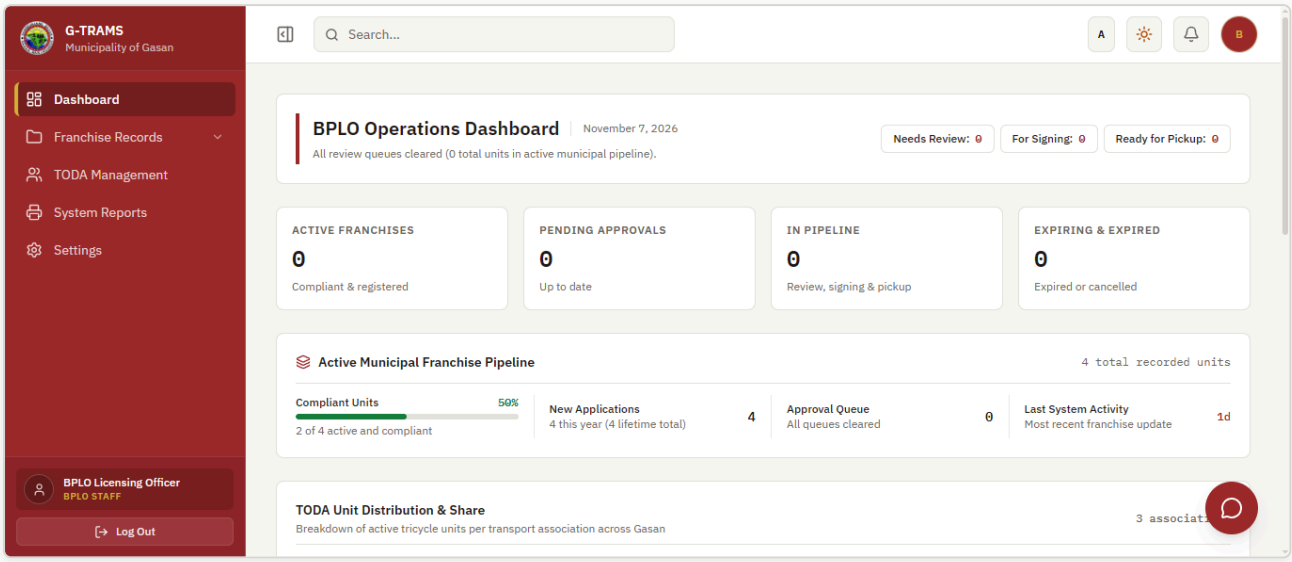


Figure 11: BPLO Operations Dashboard & Executive KPI Telemetry — Central licensing control center showing live operational metrics (Active Franchises, Pending Approvals, In Pipeline, Expiring Units), compliance pipeline distribution (50% compliant units), TODA fleet charts, and daily application intake queues.

6.2 Dual-Pane Document Review Console & AI OCR Discrepancy Checks

Clicking "Review" on any application opens the high-resolution Dual-Pane Review Console:

- **Left Canvas (Document Inspection):** High-resolution viewer with zoom, rotation (90° CW/CCW), brightness (100%–175%), contrast (100%–150%), and pan controls for scrutinizing LTO OR/CR, Cedula, Driver's License, and clearances.
- **Right Panel (AI OCR Cross-Check):** Side-by-side comparison highlighting matching data with green badges and mismatches with amber discrepancy tags. Flags third-party CR ownership requiring a Deed of Sale.
- **Concurrency Lock:** Real-time WebSocket locks prevent two officers from modifying the same record simultaneously.

6.3 Multi-Stage Franchise Approval Pipeline

Applications transition through discrete operational queues: `Needs Review` $ ightarrow$ `For Payment` (sent to Cashier) $ ightarrow$ `For Signing` (Mayor/MTFRB) $ ightarrow$ `Ready for Pickup` $ ightarrow$ `Active`. Every action includes a 10-second undo toast countdown.

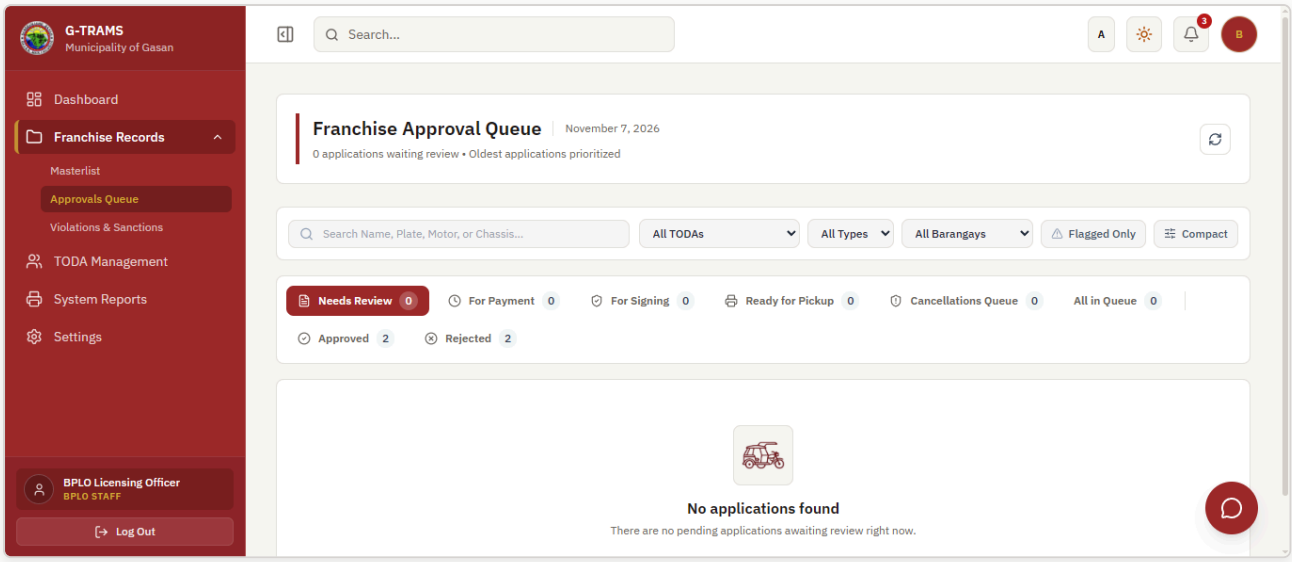


Figure 12: BPLO Multi-Stage Franchise Approval & Evaluation Queue — Structured evaluation pipeline displaying segregated workflow tabs (Needs Review, For Payment, For Signing, Ready for Pickup, Cancellations Queue, Approved, Rejected), multi-parameter filters (All TODAs, Types, Barangays, ⚠ Flagged Only), and FIFO sorting.

6.4 Progressive Traffic Sanctions & Disciplinary Enforcement

Through `/manage-revocations`, officers log traffic citations with automated penalty progression:

- **1st Offense:** Official Warning $ ightarrow$ Status: `Active - Under Warning`.
- **2nd Offense:** 15-day suspension + ₱500 municipal fine $ ightarrow$ Status: `Suspended`.
- **3rd Offense:** Permanent Franchise Revocation $ ightarrow$ Status: `Revoked`.
- **Grave Offense (Colorum):** Direct Revocation on first occurrence.

6.5 Municipal Franchise Masterlist Registry & PDF Export

The Masterlist Registry (`/franchise-masterlist`) indexes all active, expired, and historical records. Officers can execute batch MTOP certificate printing and export the full registry to PDF.

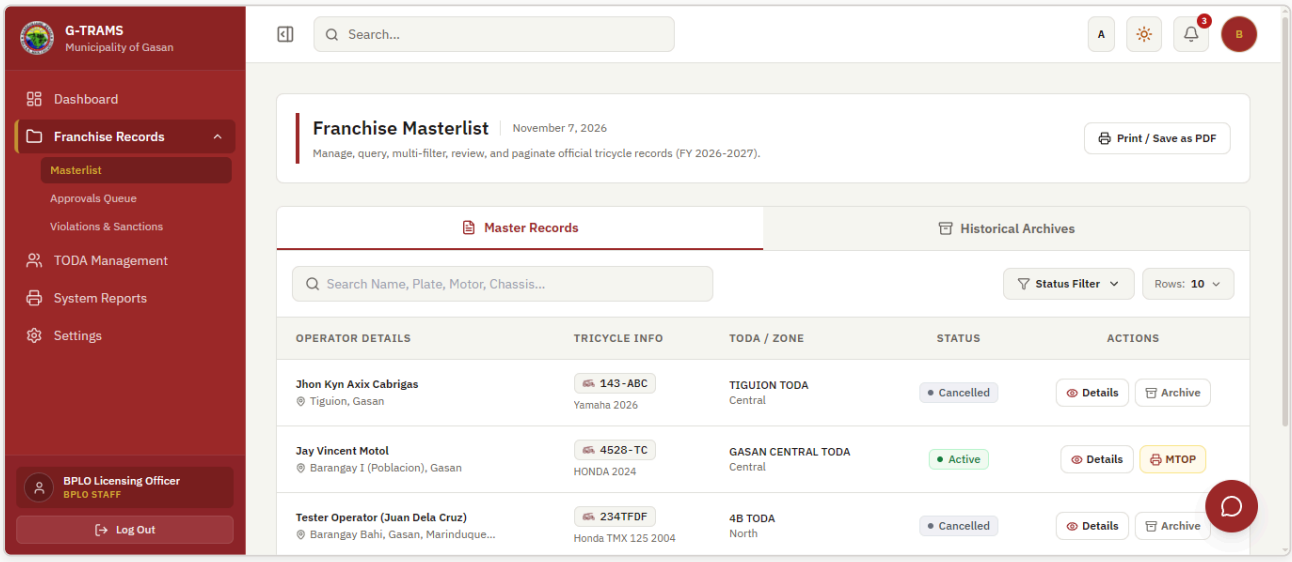


Figure 13: Municipal Franchise Masterlist Registry & Historical Archive — Official municipal registry detailing verified franchise units, plate numbers (143-ABC, 4528-TC), affiliated operators (Jhon Kyn Axix Cabrigas, Jay Vincent Motol), TODA assignments, status tags (Active, Cancelled), and one-click "Print / Save as PDF" export button.

7. MUNICIPAL CASHIER GUIDE

The Cashier Portal is an isolated financial terminal restricted to Municipal Treasury disbursing officers for regulatory fee collection and COA audit compliance.

7.1 Isolated Treasury Terminal & Security Controls

Operating at Treasury Counter 2 (`/cashier-dashboard`), the terminal displays live financial metrics: Awaiting Collection, Today's Total Collections (PHP), and Total Processed Transactions.

7.2 Over-the-Counter Cash Collection & Fee Schedule

The cashier verifies the assessed regulatory fees: Basic Franchise Fee (₱500.00), Sticker/Inspection Fee (₱150.00), and late monthly renewal penalties (₱50.00/month). Payments are collected in Cash Over-the-Counter.

7.3 Mandatory Physical Official Receipt (OR) Serialization

In compliance with COA regulations, the cashier issues a physical paper receipt from the municipal booklet and enters the serialized number:
`OR-GASAN-2026-XXXX` (e.g. `OR-GASAN-2026-0042`).
The system enforces uppercase formatting and uniqueness to prevent duplicate receipt accounting.

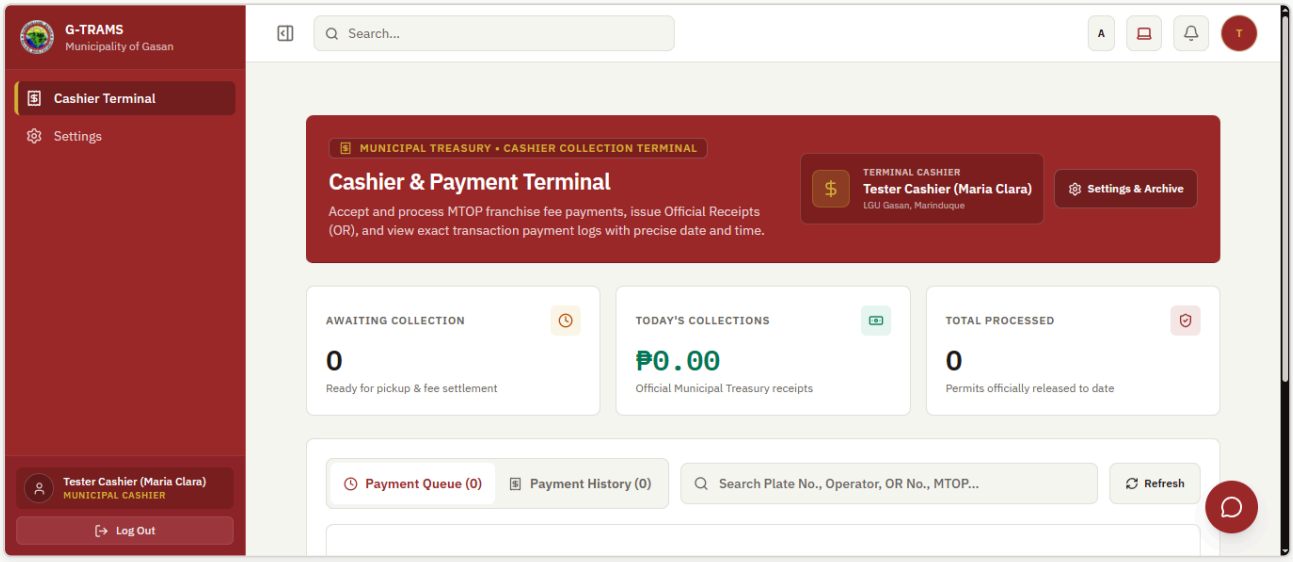


Figure 14: Municipal Treasury Cashier Collection Terminal — Isolated financial collection console showing real-time revenue metric cards ("Awaiting Collection", "Today's Collections", "Total Processed"), searchable payment queue, mandatory physical Official Receipt booklet serialization (OR-GASAN-2026-XXXX), and daily COA remittance reconciliation.

7.4 COA Daily Remittance CSV Export & Turnover Voucher Printing

Procedure 7.1: Daily Treasury Revenue Reconciliation

Purpose: Generate COA-compliant daily financial turnover sheets and accounting export at shift close (5:00 PM).

1. Click **"Daily Remittance Sheet"** on the Cashier Terminal action bar.
2. Review the summary breakdown: total transactions, cash receipts, and total municipal turnover.
3. Click **"Export Remittance CSV"** to download: `GTRAMS_Treasury_Daily_Remittance_YYYY-MM-DD.csv`.
4. Click **"Print Turnover Voucher"** to print the audit turnover sheet complete with signature blocks for the Collecting Cashier and Municipal Treasurer.
5. Submit physical cash collections and signed vouchers to the Treasury vault.

✓ **Expected Result: Treasury reconciliation is completed in under 1 minute with 100% audit accuracy.**

8. SYSTEM ADMINISTRATOR & IT OPERATIONS GUIDE

The IT Administrator module governs user directory security, system telemetry, global ordinance configurations, and automated disaster recovery protocols.

8.1 IT Diagnostics & Server Telemetry Monitoring

The IT Dashboard (`/admin-dashboard`) monitors Node.js process runtime uptime, heap memory utilization, MongoDB Atlas ping latency in milliseconds, cluster connection state, and live security audit logs.

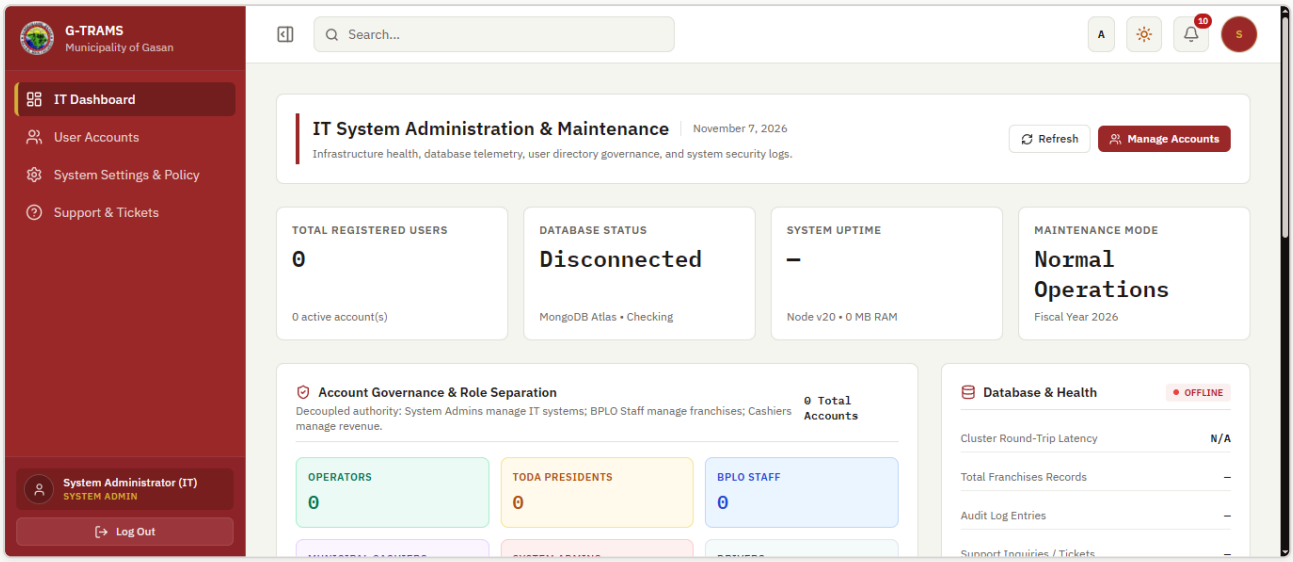


Figure 15: System Administrator IT Diagnostics & Server Telemetry Dashboard — Municipal IT oversight console displaying server status, MongoDB Atlas round-trip latency, Node.js runtime process memory allocation, system uptime counter, maintenance mode toggle, and live security audit event stream.

8.2 User Accounts Directory & Role-Based Access Control (RBAC) Governance

Through `/user-management`, IT staff oversee user accounts across all 6 tiers:

- **Socket Presence Tracking:** Displays real-time status: • `1 Online Now`, `Active Xm ago`, `Offline`.
- **Protected Admin Guard:** Administrator accounts are tagged `System Admin (You) Protected`, disabling deactivation buttons to prevent accidental self-lockout.
- **Account Deactivation & Appeals:** Suspends compromised accounts with mandatory justification notes.

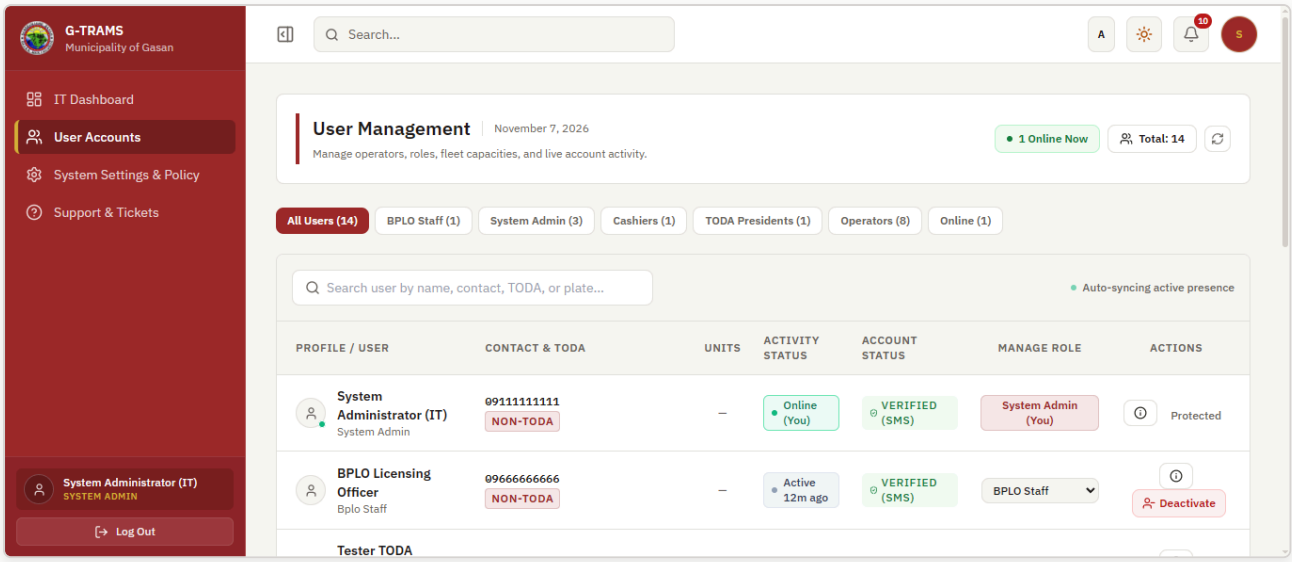


Figure 16: User Accounts Directory & Role-Based Access Control (RBAC) Governance — Administrative user directory displaying real-time socket presence tracking (• 1 Online Now, Active 12m ago), role distribution filters across 6 user tiers, SMS verification badges, and account protection guardrails ((You) Protected).

8.3 System Policy & Fee Schedule Configuration

Under Admin Settings (`/admin/settings`), administrators configure municipal parameters: Franchise Validity (New: 3 Years, Renewal: 1 Year, Expiry Warning: 30 Days), Fiscal Year, Franchise Fee (₱500.00), Base Fare (₱15.00), Per-Km Rate (₱2.50), and the System Maintenance Mode toggle.

8.4 Disaster Recovery Protocol (DRP): Weekly Backups, Offline Snapshots & Database Restore

G-TRAMS establishes a three-tier Disaster Recovery Protocol (DRP):

- **Automated Weekly Cron Archive:** Backs up all 13 MongoDB collections every **Sunday at 12:00 MN (Midnight)** into compressed `.json.gz` archives with rolling 8-week retention.
- **1-Click Offline Snapshot:** Direct download of compressed gzip database snapshots to encrypted local USB drives for offsite physical custody.
- **Safe Restoration Engine:** Full database restoration requiring typing the explicit uppercase safety phrase: `RESTORE`. Restores all 13 collections with verified data integrity in seconds.

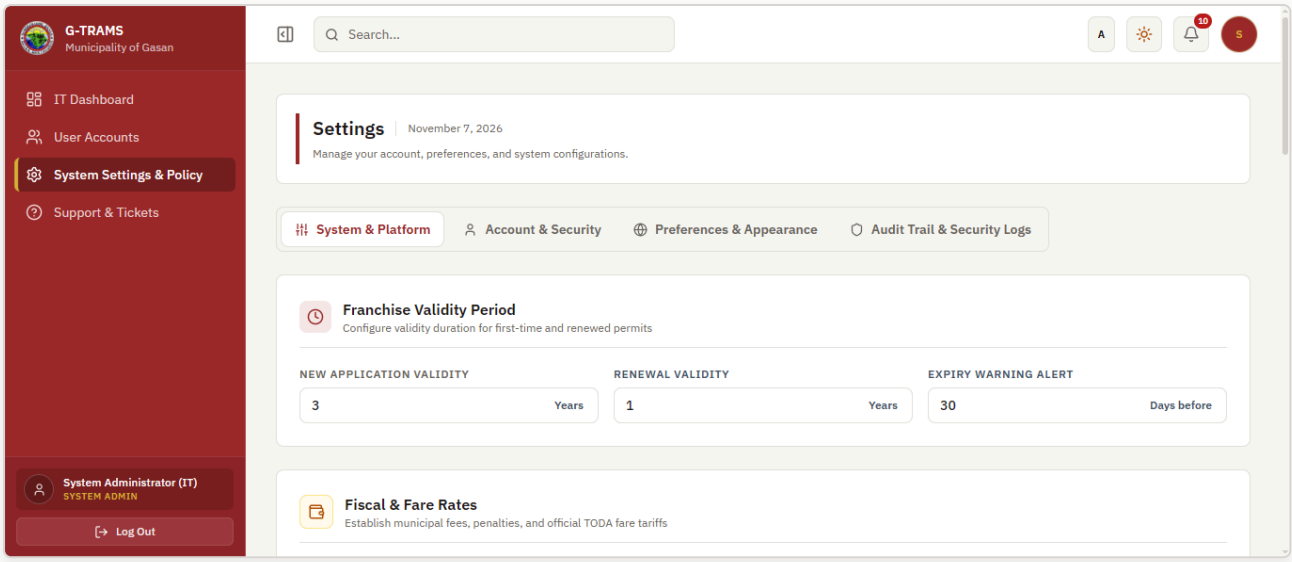


Figure 17: System Policy Configuration & Disaster Recovery Protocol (DRP) — System settings console detailing franchise validity durations (New: 3 Years, Renewal: 1 Year, 30-day expiry warning alert), municipal fare tariff settings, automated Sunday 12:00 MN cron backup status, 1-click .json.gz offline snapshot download, and full 13-collection restore requiring safety confirmation phrase "RESTORE".

9. TROUBLESHOOTING, FAQS & SUPPORT DIRECTORY

The following diagnostic matrix provides verified resolutions for common operational symptoms:

9.1 Error Resolution Matrix

SYMPTOM / ERROR MESSAGE	PROBABLE ROOT CAUSE	VERIFIED STEP-BY-STEP RESOLUTION
"Invalid credentials or incorrect password"	Typo in password or account not yet activated via OTP.	Verify Caps Lock is disabled. Ensure you are using an 11-digit mobile number (09xxxxxxxxx) or verified email. Click "Forgot Password?" to reset.
"OTP code has expired or is invalid"	More than 5 minutes elapsed since OTP dispatch.	Click "Resend OTP". Ensure active mobile cellular reception in Gasan and enter the newly dispatched 6-digit code immediately.
"Document upload failed / File exceeds size limit"	Unsupported format or image exceeds 10 MB limit.	Upload files in .jpg, .png, or .pdf format. Crop margins or reduce camera photo resolution if image exceeds 10 MB.
"Duplicate vehicle identifier detected"	Plate, Engine, or Chassis number already registered.	Cross-check LTO CR. If acquired second-hand, confirm the previous owner filed a formal Dropping / Transfer at BPLO.
"Unit Capacity 2/2 Reached — Quota Ceiling Enforced"	Operator already holds two active franchises.	Under Section 4 of Municipal Franchising Ordinance, operators are limited to 2 units. You must drop an existing unit before filing a new one.
"Notice: Official TODA Certificate Required"	Applicant name is unlisted in association roster.	Consult TODA President to encode membership in digital roster. If unlisted, upload a signed physical TODA Clearance certificate.
"Access Denied (403 Unauthorized)"	User role lacks permissions for the target module.	Log out and authenticate with the authorized account (e.g., Cashier for treasury, BPLO for approvals, Admin for IT).
"Official Receipt (OR) Number already exists"	Cashier entered an OR number already recorded.	Inspect physical municipal receipt booklet to confirm accurate sequential serial number. Duplicate entries are blocked for COA compliance.
"Restore Aborted: Invalid confirmation phrase"	Administrator failed to type safety phrase.	Type the exact word RESTORE in all capital letters in the confirmation prompt to authorize database restoration.
"Server connection timeout / Maintenance in progress"	Scheduled maintenance mode or temporary network drop.	Check internet connection. If maintenance mode is active, wait until municipal IT staff complete scheduled upgrades.

9.2 Frequently Asked Questions (FAQs)

Q1: What is the processing turnaround time for an MTOP application in G-TRAMS?

Under RA 11032 (Ease of Doing Business), complete applications with verified credentials are processed within **2 to 3 business days** from online submission to BPLO inspection, payment, and final permit release.

Q2: When am I eligible to renew my commercial tricycle franchise?

Franchise permits are valid for **1 year for Renewals** and **3 years for New Franchises**. The system triggers an alert on your dashboard **30 days prior** to expiration, activating early renewal filing.

Q3: What should I do if my primary mobile phone is lost or damaged?

During registration, G-TRAMS captures a mandatory **Secondary Recovery Contact**. Click "Forgot Password?", enter your recovery contact, and receive an OTP to regain account access without visiting municipal hall.

Q4: How does a traffic enforcer know my digital e-Copy MTOP is authentic?

Every digital MTOP certificate and physical sidecar sticker features a **dynamic encrypted QR code**. Scanning it retrieves live, tamper-evident records directly from the municipal database at `/verify/:id`.

Q5: Is my personal document data protected under Philippine privacy laws?

Yes. In compliance with the **Data Privacy Act of 2012 (RA 10173)**, passwords use cryptographic bcrypt hashing, biometric markers like blood type are excluded from AI OCR extraction, and documents are stored in secure cloud vaults with expiring tokens.

9.3 LGU Office Directory & Official Support Channels

DEPARTMENT / OFFICE	PHYSICAL OFFICE LOCATION	OPERATING HOURS	CONTACT NUMBER & EMAIL
Office of the Municipal Mayor	2nd Floor, Gasan Municipal Hall, Poblacion, Gasan	Mon – Fri, 8:00 AM – 5:00 PM	(042) 342-1001 • mayor@gasan.ph
BPLO Franchising Division	2nd Floor, Gasan Municipal Hall, Poblacion, Gasan	Mon – Fri, 8:00 AM – 5:00 PM	(042) 342-1234 • bplo@gasan.ph
Municipal Treasury (Counter 2)	Ground Floor, Gasan Municipal Hall, Poblacion, Gasan	Mon – Fri, 8:00 AM – 5:00 PM	(042) 342-1235 • treasury@gasan.ph
Traffic Enforcement / CTMO	Ground Floor (East Wing), Gasan Municipal Hall	24/7 Roadside Operations	(042) 342-1111 • traffic@gasan.ph
G-TRAMS IT Technical Support	College of Information and Computing Sciences, MarSU	Mon – Fri, 8:00 AM – 5:00 PM	support@gtrams.ph • cics@mar-su.edu.ph

— End of Deliverable #7: User's Manual of the System —

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